

```
In [1]: import pandas as pd
```

READ DATASET

```
In [2]: movies = pd.read_csv(r'D:\movie.csv')
```

```
In [3]: movies
```

```
Out[3]:
```

	movieId	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	2	Jumanji (1995)	Adventure Children Fantasy
2	3	Grumpier Old Men (1995)	Comedy Romance
3	4	Waiting to Exhale (1995)	Comedy Drama Romance
4	5	Father of the Bride Part II (1995)	Comedy
...	...	...	...
27273	131254	Kein Bund für's Leben (2007)	Comedy
27274	131256	Feuer, Eis & Dosenbier (2002)	Comedy
27275	131258	The Pirates (2014)	Adventure
27276	131260	Rentun Ruusu (2001)	(no genres listed)
27277	131262	Innocence (2014)	Adventure Fantasy Horror

27278 rows × 3 columns

```
In [8]: movies.shape
```

```
Out[8]: (27278, 3)
```

```
In [9]: movies.head(1)
```

```
Out[9]:
```

	movieId	title	genres
0	1	Toy Story (1995)	Adventure Animation Children Comedy Fantasy

```
In [10]: movies.columns
```

```
Out[10]: Index(['movieId', 'title', 'genres'], dtype='object')
```

```
In [11]: tags = pd.read_csv(r'D:\tag.csv')
```

In [12]: `tags`

Out[12]:

	userId	movieId	tag	timestamp
0	18	4141	Mark Waters	24-04-09 18:19
1	65	208	dark hero	10-05-13 1:41
2	65	353	dark hero	10-05-13 1:41
3	65	521	noir thriller	10-05-13 1:39
4	65	592	dark hero	10-05-13 1:41
...	...	...	...	...
89214	22338	64034	World War II	16-11-14 17:43
89215	22338	83182	skits	16-11-14 17:10
89216	22338	83182	surreal	16-11-14 17:10
89217	22338	85179	anime	16-11-14 17:21
89218	22338	85179	hacking	2014-

89219 rows × 4 columns

In [13]: `tags.shape`

Out[13]: (89219, 4)

In [14]: `tags.columns`

Out[14]: Index(['userId', 'movieId', 'tag', 'timestamp'], dtype='object')

In [15]: `ratings = pd.read_csv(r'D:\rating.csv')`

In [16]: `ratings`

Out[16]:

	userId	movieId	rating	timestamp
0	1	2	3.5	02-04-05 23:53
1	1	29	3.5	02-04-05 23:31
2	1	32	3.5	02-04-05 23:33
3	1	47	3.5	02-04-05 23:32
4	1	50	3.5	02-04-05 23:29
...	...	...	...	...
1048570	7120	168	5.0	02-04-07 19:44
1048571	7120	253	4.0	02-04-07 19:30
1048572	7120	260	5.0	02-04-07 19:27
1048573	7120	261	4.0	02-04-07 19:49
1048574	7120	266	3.5	02-04-07 19:34

1048575 rows × 4 columns

In [17]: `ratings.shape`

Out[17]: (1048575, 4)

In [18]: `ratings.columns`

Out[18]: Index(['userId', 'movieId', 'rating', 'timestamp'], dtype='object')

In [19]: `del ratings['timestamp']`
`del tags['timestamp']`In [20]: `ratings.head()`

Out[20]:

	userId	movieId	rating
0	1	2	3.5
1	1	29	3.5
2	1	32	3.5
3	1	47	3.5
4	1	50	3.5

In [21]: `tags.head()`

```
Out[21]:
```

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

DATA STRUCTURE

```
In [22]: row_0 = tags.iloc[0]
         type(row_0)
```

```
Out[22]: pandas.core.series.Series
```

```
In [23]: tags.iloc[0]
```

```
Out[23]: userId          18
         movieId       4141
         tag           Mark Waters
         Name: 0, dtype: object
```

```
In [24]: row_0.index
```

```
Out[24]: Index(['userId', 'movieId', 'tag'], dtype='object')
```

```
In [25]: row_0['userId']
```

```
Out[25]: np.int64(18)
```

```
In [26]: 'rating' in row_0
```

```
Out[26]: False
```

```
In [27]: row_0.name
```

```
Out[27]: 0
```

```
In [28]: row_0 = row_0.rename('firstRow')
         row_0.name
```

```
Out[28]: 'firstRow'
```

DATA FRAMES

```
In [29]: tags.head()
```

Out[29]:

	userId	movieId	tag
0	18	4141	Mark Waters
1	65	208	dark hero
2	65	353	dark hero
3	65	521	noir thriller
4	65	592	dark hero

In [30]: `tags.index`

Out[30]: `RangeIndex(start=0, stop=89219, step=1)`

In [31]: `tags.columns`

Out[31]: `Index(['userId', 'movieId', 'tag'], dtype='object')`

In [32]: `tags.iloc[ [0,11,500] ]`

Out[32]:

	userId	movieId	tag
0	18	4141	Mark Waters
11	65	1783	noir thriller
500	342	55908	entirely dialogue

DESCRIPTIVE STATISTICS

In [33]: `ratings['rating'].describe()`

Out[33]:

count	1.048575e+06
mean	3.529272e+00
std	1.051919e+00
min	5.000000e-01
25%	3.000000e+00
50%	4.000000e+00
75%	4.000000e+00
max	5.000000e+00
Name: rating, dtype: float64	

In [34]: `ratings.describe()`

Out[34]:

	userId	movieId	rating
count	1.048575e+06	1.048575e+06	1.048575e+06
mean	3.527086e+03	8.648988e+03	3.529272e+00
std	2.018424e+03	1.910014e+04	1.051919e+00
min	1.000000e+00	1.000000e+00	5.000000e-01
25%	1.813000e+03	9.030000e+02	3.000000e+00
50%	3.540000e+03	2.143000e+03	4.000000e+00
75%	5.233000e+03	4.641000e+03	4.000000e+00
max	7.120000e+03	1.306420e+05	5.000000e+00

In [35]: `ratings['rating'].mean()`

Out[35]: `np.float64(3.5292716305462175)`

In [36]: `ratings.mean()`

Out[36]:

```

userId      3527.086123
movieId     8648.988281
rating       3.529272
dtype: float64

```

In [37]: `ratings['rating'].min()`

Out[37]: `0.5`

In [38]: `ratings['rating'].max()`

Out[38]: `5.0`

In [39]: `ratings['rating'].std()`

Out[39]: `1.0519187535891295`

In [40]: `ratings['rating'].mode()`

Out[40]:

```

0      4.0
Name: rating, dtype: float64

```

In [41]: `ratings.corr()`

Out[41]:

	userId	movieId	rating
userId	1.000000	-0.002837	0.017105
movieId	-0.002837	1.000000	0.002550
rating	0.017105	0.002550	1.000000

```
In [42]: filter1 = ratings['rating'] > 10
print(filter1)
filter1.any()
```

```
0      False
1      False
2      False
3      False
4      False
...
1048570 False
1048571 False
1048572 False
1048573 False
1048574 False
Name: rating, Length: 1048575, dtype: bool
```

Out[42]: np.False_

```
In [43]: filter2 = ratings['rating'] > 0
filter2.all()
```

Out[43]: np.True_

DATA CLEANING :HANDING MISSING DATA

```
In [44]: movies.shape
```

Out[44]: (27278, 3)

```
In [45]: movies.isnull().any().any()
```

Out[45]: np.False_

```
In [46]: ratings.shape
```

Out[46]: (1048575, 3)

```
In [47]: ratings.isnull().any().any()
```

Out[47]: np.False_

```
In [48]: tags.shape
```

```
Out[48]: (89219, 3)
```

```
In [49]: tags.isnull().any().any()
```

```
Out[49]: np.False_
```

```
In [50]: tags=tags.dropna()
```

```
In [51]: tags.isnull().any().any()
```

```
Out[51]: np.False_
```

```
In [52]: tags.shape
```

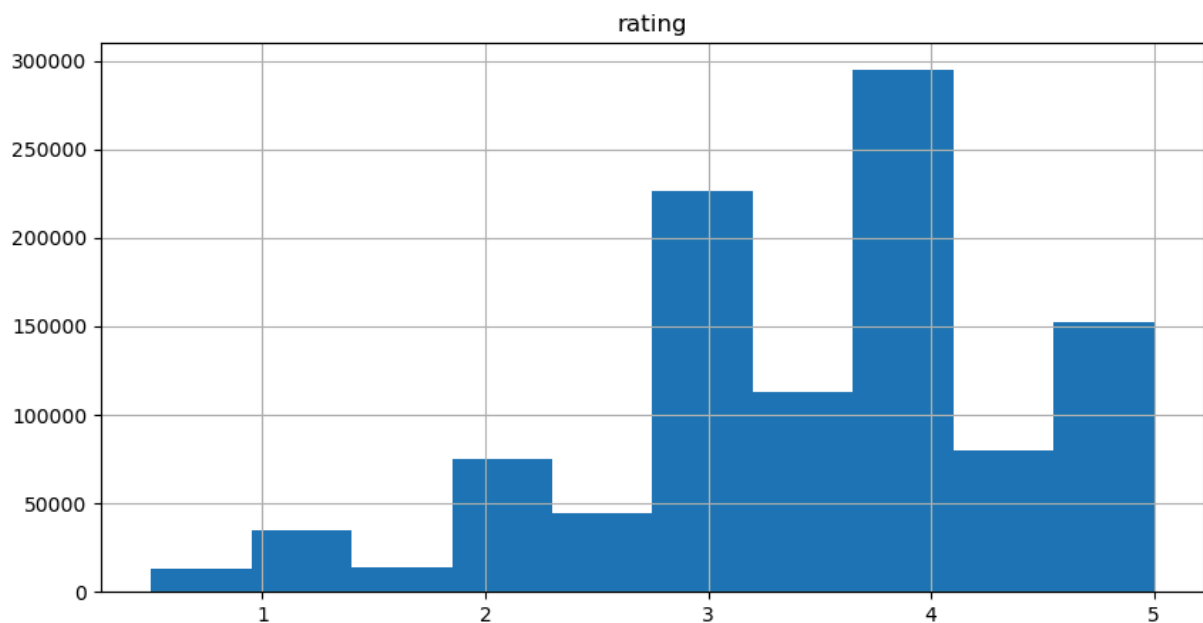
```
Out[52]: (89219, 3)
```

DATA VISUALIZATION

```
In [53]: %matplotlib inline
```

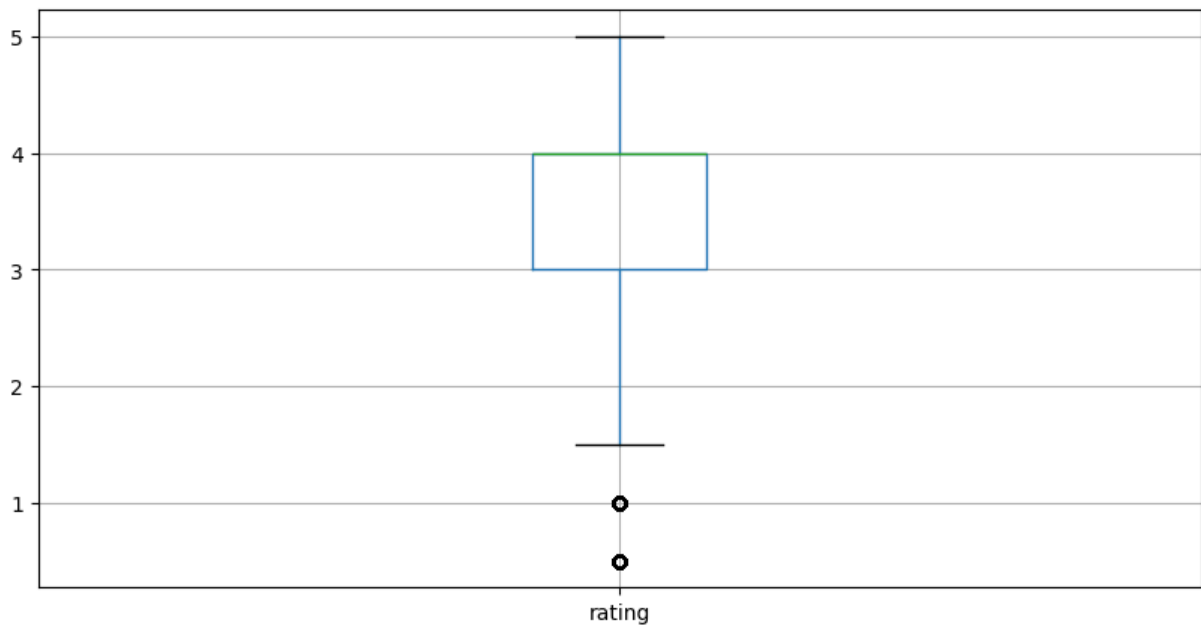
```
ratings.hist(column='rating', figsize=(10,5))
```

```
Out[53]: array([[<Axes: title={'center': 'rating'}>]], dtype=object)
```



```
In [54]: ratings.boxplot(column='rating', figsize=(10,5))
```

```
Out[54]: <Axes: >
```



SLICING OUT COLUMNS

```
In [55]: tags['tag'].head()
```

```
Out[55]: 0    Mark Waters
          1    dark hero
          2    dark hero
          3    noir thriller
          4    dark hero
          Name: tag, dtype: object
```

```
In [56]: movies[['title', 'genres']].head()
```

```
Out[56]:
```

	title	genres
0	Toy Story (1995)	Adventure Animation Children Comedy Fantasy
1	Jumanji (1995)	Adventure Children Fantasy
2	Grumpier Old Men (1995)	Comedy Romance
3	Waiting to Exhale (1995)	Comedy Drama Romance
4	Father of the Bride Part II (1995)	Comedy

```
In [57]: ratings[-10:]
```

Out[57]:

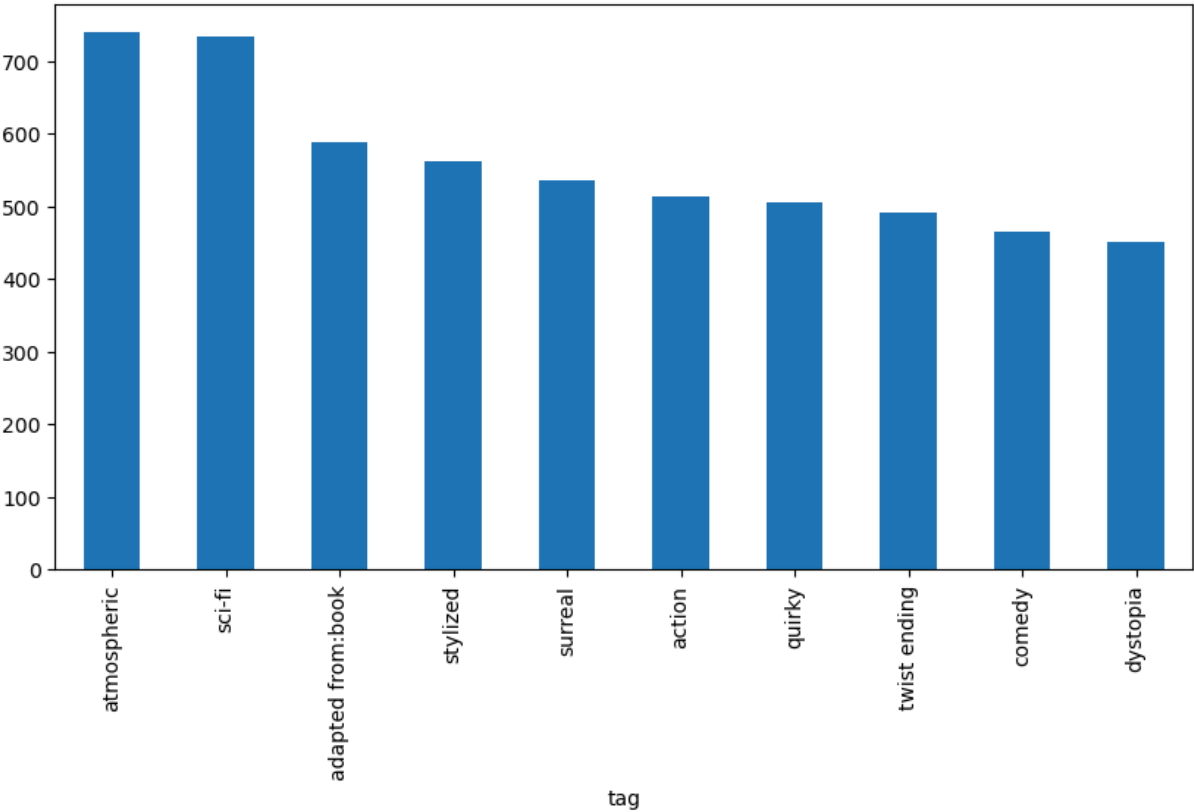
	userId	movieId	rating
1048565	7120	141	5.0
1048566	7120	151	5.0
1048567	7120	153	0.5
1048568	7120	161	4.0
1048569	7120	163	4.5
1048570	7120	168	5.0
1048571	7120	253	4.0
1048572	7120	260	5.0
1048573	7120	261	4.0
1048574	7120	266	3.5

```
In [58]: tag_counts = tags['tag'].value_counts()
tag_counts[-10:]
```

```
Out[58]: tag
jeff                      1
flying jokes              1
Avoid for your sanity     1
No Shadows                1
No mystery                1
No Game                   1
Action Movie              1
Batman voice              1
Bane's voice              1
skits                     1
Name: count, dtype: int64
```

```
In [59]: tag_counts[:10].plot(kind='bar', figsize=(10,5))
```

```
Out[59]: <Axes: xlabel='tag'>
```



```
In [ ]:
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```